



AI-Powered Intelligent Job Search and Career System

Data Science and AI Project

Milestone 2 Report

Group 4

Prepared By:

Gaurav Kumar (22f1001105)
Dev Gupta (22f2000888)
Abhinav Ohri (24f1002064)
Pranav N (22f2000117)
Praveena N (22f3001454)

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1. Introduction

The project, **AI-Powered Intelligent Job Search and Career System** aim to support candidates during the job search process by connecting resume parsing, candidate profile creation, career recommendation, and job discovery/matching into one structured workflow. In Milestone 1, the project scope was finalized around three main modules: **Resume Parsing**, **Career Recommendation**, and **Job Discovery and Job Matching**. The system first extracts structured information from a resume, then uses that profile to recommend suitable career roles, and finally ranks relevant jobs based on profile fit.

The objective of **Milestone 2** is to identify, understand, clean, preprocess, and prepare datasets for these modules.

2. Dataset Identification

2.1 Dataset List

Module	Dataset Name	Source	Type of dataset	Purpose
Resume Parsing	Resume Data (PDF)	Kaggle — Hadikp/resume-data-pdf	Publicly accessible Kaggle dataset; licence and permitted academic use verified	Provide real resume documents for document-quality analysis, multimodal parsing, structured field extraction, and gold-set evaluation
Career Recommendation	ESCO Dataset v1.2.1 — English CSV classification	ESCO Official Portal	Public dataset	Map candidate skills and experience to suitable career roles
Job Discovery and Matching	Candidate Job Role Dataset and LinkedIn Job Postings Dataset	Kaggle	Public datasets	Provide candidate profiles and real-world job postings for semantic matching, feature engineering, ranking, and evaluation

2.2 Resume Parsing Dataset

The Resume Parsing module uses the **Resume Data (PDF)** dataset published on Kaggle. It contains 8,905 resume documents originally organised into **97 occupation-labelled** folders, which map to **43 canonical job categories** with no documents removed. The dataset was selected because it:

- contains complete resume documents which allows us to evaluate the parser on real layout, image quality, and readability.
- has broad occupational coverage across technical, administrative, management, design, finance, healthcare, and other roles.
- is large enough to build a category-balanced evaluation set (43 categories).
- consists entirely of single-page, image-only PDFs , evaluating the hardest case, since docx extract deterministically in Python while the vision path is where accuracy is at risk.

Although the dataset covers only image-only PDFs, the production parser is built to handle every common upload format:

Uploaded format	Processing route
Selectable-text PDF	PyMuPDF text extraction → schema-constrained LLM extraction
DOCX	python-docx text extraction → schema-constrained LLM extraction
Image-only PDF or scan	Render pages → multimodal vision model
Multi-page resume	Parse each page → merge and deduplicate extracted fields
Tall scanned image	Divide into overlapping segments → parse each segment → merge results

PyMuPDF extracts the page text, and if fewer than 100 characters are recovered the document is treated as image-only and sent to the vision path, otherwise it takes the text route. A vision retry is triggered if the text route returns sparse or low-quality text.

2.2.1 Dataset Limitations and Mitigation

Limitation	Effect on evaluation	Mitigation
Single-page resumes	Multi-page merging is not tested	Parse each page and merge results
No DOCX files	Word-document parsing is not tested	Extract text using python-docx
Clean Scans (Limited noise)	May not represent blurry or photographed resumes	Test blur, rotation, compression, and low-resolution images
Class Imbalance	Large categories may dominate evaluation	Category-balanced gold set + per-category metrics

2.3 Career Recommendation Dataset

The Career Recommendation module uses the **ESCO v1.2.1 English classification dataset**. ESCO provides a structured taxonomy of occupations, skills, knowledge, competencies, alternative job titles, descriptions, and occupation–skill relationships.

The prepared dataset contains **3,039 occupations**, **13,939 cleaned skills**, and **126,051 valid occupation–skill relations**. It covers all ten major ISCO occupational groups, including managers, professionals, technicians, clerical workers, service workers, agricultural workers, craft workers, machine operators, and elementary occupations. The large number of occupations and skills provides broad occupational coverage and supports recommendation across different industries.

ESCO was developed mainly for the European labour market. Therefore, some occupation titles, regulatory descriptions, educational terminology, and alternative labels may be Europe-oriented. The system does not assume that every Indian or non-European job title has an exact ESCO equivalent.

The Career Recommendation module uses ESCO as a general reference taxonomy. Generalisation is supported through:

- occupation descriptions rather than title matching alone;
- preferred and alternative occupation titles;
- complete candidate skill and experience context;
- semantic embeddings;
- skill-specificity weighting;
- evaluation using local and non-European title variants;
- retaining uncertain mappings as unmapped rather than forcing an incorrect occupation.

This module uses semantic retrieval rather than training a classification model directly on the ESCO occupations. Therefore, generalisation is evaluated using unseen candidate profiles, paraphrased occupation titles, local job-title variations, and different combinations of skills and experience.

2.4 Job Discovery and Matching Dataset

The Job Discovery and Matching module uses two complementary datasets:

1. **Candidate Job Role Dataset** — structured candidate profiles containing skills, qualifications, experience levels, and target job roles.
2. **LinkedIn Job Postings Dataset** — real-world job postings containing titles, descriptions, companies, locations, experience requirements, employment types, posting dates, and salary information where available.

The candidate profile represents the search query, while each job posting represents a document to be retrieved and ranked.

2.4.1 Dataset suitability

The Candidate Job Role Dataset contains 1,000 structured candidate profiles, 98 unique skill combinations, 17 qualification categories, four experience levels, and 22 target job roles. It provides a controlled candidate-side dataset suitable for testing skill, qualification, role, and experience matching.

The LinkedIn dataset contains 123,849 job postings from 24,428 companies, with 72,521 unique job titles and 8,526 locations. The large number of companies, titles, and locations

provides enough variation for testing retrieval and ranking behaviour across different occupations.

2.4.2 Geographic coverage

The job dataset is not geographically balanced. Only **742 of 123,849 postings, approximately 0.60%, explicitly mention India**, while the most frequent locations are from the United States.

Therefore, the dataset is suitable for evaluating the general mechanics of the ranking system, such as semantic similarity, experience matching, and title alignment. It is not sufficient for making claims about the Indian labour market, India-specific salaries, or the availability of jobs in India.

For production deployment, live job discovery will collect permitted public postings relevant to the candidate's selected location. The offline LinkedIn dataset is used only for development and evaluation of the matching pipeline.

2.4.3 Recency

The postings were listed between **24 March 2024 and 20 April 2024**, covering a 26-day period. This provides a consistent and relatively recent snapshot for ranking experiments, although it does not represent seasonal or long-term changes in hiring demand.

2.4.4 Industry coverage

The most frequent titles include Sales Manager, Customer Service Representative, Project Manager, Administrative Assistant, accounting roles, healthcare roles, and software-related roles. Therefore, the job dataset is multi-industry rather than IT-only.

However, the candidate dataset is mainly focused on software and technology roles. This creates a candidate-job industry mismatch. Evaluation is therefore limited to job postings that can reasonably be aligned with the 22 candidate-role categories.

2.4.5 ESCO alignment

A preliminary ESCO alignment test was performed using occupation titles, alternative ESCO titles, exact matching, and token-based fuzzy matching.

- **68.2% of the 22 candidate job roles** received an ESCO match.
- **92% of the 50 most frequent LinkedIn job titles** received an exact or fuzzy ESCO match.

These values represent mapping coverage, not guaranteed mapping accuracy. Exact matches are accepted directly, while low-confidence fuzzy matches are reviewed because short or ambiguous titles may be mapped incorrectly. Unreliable mappings remain unmapped rather than being forced into an unrelated ESCO occupation.

2.4.6 Job Discovery scope

The static LinkedIn dataset supports offline Job Matching experiments. Live Job Discovery is handled separately during application development using permitted public pages or SearXNG-based search.

Retrieved jobs are converted into the same structured schema and stored in the database so that:

- repeated web requests are reduced;
- jobs can be deduplicated;
- posting sources and collection times can be recorded;
- retrieved jobs can pass through the same ranking pipeline as offline jobs.

2.5 Alternatives Considered

Alternative	Reason Considered	Reason Not Used as Main Dataset
O*NET	Well-known occupation and skill database	ESCO provides direct occupation-skill relation files
Snehaan Bhawal Resume Dataset	Contains more than 2,400 resume PDFs and category labels	Provides fewer job categories and less document diversity
ResumeAtlas	Larger dataset with 13,000+ resume records	Mainly provides OCR-extracted text and does not preserve the original resume document layout

3. Dataset Description

3.1 Resume Parsing Dataset

Item	Details
Dataset name	Resume Data (PDF)
Source	Kaggle — hadikp/resume-data-pdf
Raw records	8,905 resume PDFs
Format	Single-page, image-only PDF documents
Raw category folders	97
Canonical categories	43
Corrupt or unreadable files	0
Files with extractable PDF text	0
Exact duplicate files	0
Perceptual groups	6,611

Gold evaluation PDFs	86
Development set	34
Test set	52
Input feature	Rendered resume-page image
Weak label	Canonical resume category
Target fields	Contact, skills, education, experience, projects, certifications, and job titles

3.1.2 Resume Parsing target schema

Target group	Extracted fields
contact	Name, location, and public professional links.
skills	Skills, software tools, technologies, and competencies
education	Degree, field, institution, start year, and end year
experience	Job title, company, location, dates, current-role flag, and description
projects	Project name, description, and technologies
certifications	Certification name, issuer, and year
job_titles	Normalised previous or current role titles

An OCR sample of 200 resumes gives a median length of ~500 tokens and a 95th-percentile of ~1,655 tokens, so nearly every resume fits in a single model request without chunking.

3.1.3 Gold Annotation Methodology

The gold evaluation dataset contains **86 unique resumes**: 43 categories × 2 resumes per category.

Selection steps:

1. Map all resumes to 43 canonical categories.
2. Detects exact duplicates (MD5) and near-duplicates (perceptual hash, Hamming ≤ 6).
3. Reduce each duplicate group to one representative (smallest file) so no two visually similar resumes enter the set.
4. Sample two representatives per category ($43 \times 2 = 86$).

All sampling and the dev/test split use fixed random seed 42.

The annotations were initially drafted using AI from the resume images. A subset of **15 resumes was reviewed by three team members** as a quality-control check. The remaining 71 records were not individually human-verified. No formal inter-annotator

agreement was calculated because the annotations were not independently created by multiple annotators.

3.1.4 Annotation Quality Assurance

Annotation quality was checked through four layers:

Quality-control layer	Validation performed
Annotation guidelines	Fixed rules for extraction, missing values, dates, lists, and unsupported information
Schema validation	All 86 records validated using <code>parsed_resume_schema.json</code>
Consistency checks	Category coverage, field counts, empty-field checks, and unusual records inspected through <code>_manifest.csv</code>
Human review	Fifteen resumes reviewed by three team members against the original images

Annotations followed a fixed guideline: information was transcribed only when visible in the resume; missing scalar fields were stored as null; missing list fields were stored as []; each employment role was stored as a separate object; dates were preserved as written; duplicate skills were removed case-insensitively; and email addresses and phone numbers were excluded.

All 43 categories are represented in the gold set. All 86 records contain at least one extracted skill. Six records contain no work-experience entries and were retained because they represent fresher or skills-focused resumes rather than annotation errors.

3.2 Career Recommendation Dataset

3.2.1 Raw Files Used

File	Purpose
<code>occupations_en.csv</code>	Contains ESCO occupation titles, alternative titles, descriptions, and occupation URIs
<code>skills_en.csv</code>	Contains ESCO skill titles, alternative skill names, descriptions, skill types, and skill URIs
<code>occupationSkillRelations_en.csv</code>	Connects occupations with related skills
<code>ISCOGroups_en.csv</code>	Provides broader occupation grouping, kept for reference
<code>skillGroups_en.csv</code>	Provides broader skill grouping, kept for reference

3.2.2 Processed Dataset Summary

Item	Value
Clean occupations	3,039

Clean skills	13,939
Clean occupation-skill relations	126,051
Final prepared career role records	3,039
Occupations with zero skills	0
Average skills per occupation	41.48
Unused skills	464

3.2.3 Final Career Recommendation Schema

Column	Description
occupation_id	Stable ESCO occupation ID extracted from concept URI
conceptUri	Original ESCO occupation URI
occupation_title	Main ESCO occupation title
alt_occupation_title	Alternative occupation titles
occ_description	ESCO occupation description
num_skills	Number of skills linked to the occupation
skills_list	Full list of linked ESCO skills
skills_for_embedding_text	Selected skills used in embedding text
career_role_context	Final combined text used for embedding generation

3.3 Job Discovery and Job Matching data

Metric	Candidate Job Role Dataset	LinkedIn Job Postings Dataset
Raw records	1,000	123,849
Raw features	5	31
Features retained	5	16, including derived datetime
Format	CSV	CSV
Data format	CSV	CSV
Native target	job_role	None
Ranking target	Pair-level relevance label	Pair-level relevance label

3.3.2 Candidate Job Role Features:

Feature	Description
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candidate_id	Unique candidate identifier
skills	Candidate's technical and professional skills
qualification	Highest educational qualification
experience_level	Entry, Mid, Senior, or Expert experience category
job_role	Candidate's current or target role

3.3.3 LinkedIn Job Postings Dataset Features:

Feature	Description
job_id	Unique job-posting identifier
title	Free-text job title
description	Job responsibilities, requirements, and qualifications
company_name	Hiring company
location	Job location
formatted_experience_level	Required job experience category
skills_desc	Structured skill description where available
formatted_work_type	Employment type, such as full-time, contract, or internship
listed_time	Posting timestamp used for freshness
normalized_salary	Normalized salary where disclosed

The candidate dataset contains 98 unique skill combinations, 17 qualification categories, four experience-level categories, and 22 job-role categories. The job dataset contains 24,428 companies, 72,521 unique job titles, and 8,526 locations.

3.3.4 Practical dataset limitations

Candidate dataset

- It contains only 1,000 candidate profiles.
- Only 98 unique skill combinations occur across the 1,000 profiles.
- The candidate roles are mainly technology-oriented.
- Candidate skills appear structured and templated rather than written like real resume text.
- Candidate location, preferred employment type, and salary preference are not available.

Practical consequence: the dataset is suitable for controlled matching experiments but does not represent the full noise and diversity of real parsed resumes. During deployment, structured profiles produced by the Resume Parsing module will replace these synthetic candidate records.

Job dataset

- It represents LinkedIn-visible jobs rather than the complete job market.
- It is mainly U.S.-oriented and contains limited Indian coverage.
- It covers only a 26-day time window.
- Job titles are extremely fragmented.
- Optional structured fields have uneven coverage and are not required for the core matching process. The job title and full description are used as the consistent matching input, while salary and structured skill fields are used only when available.
- Exact duplicate checking does not automatically detect repostings with slightly different text or locations.

The notebook identifies 72,521 unique titles, with 60,922 titles occurring only once. This confirms that raw titles cannot be treated directly as stable job classes.

4. Data Governance

Area	Resume Parsing	Career Recommendation	Job Discovery and Matching
Data Source & Licensing	Publicly accessible Kaggle dataset published under hadikp/resume-data-pdf. The dataset source, licence conditions, permitted academic use, and download date are recorded in the project documentation	ESCO official public dataset	Candidate Job Role Dataset used under the MIT licence. LinkedIn Job Postings Dataset used under CC BY-SA 4.0. Sources, versions, download dates, and attribution are documented
Ownership	Dataset published by Kaggle user hadikp; the source consists mainly of resumes produced using resume templates and resume-builder platforms	European Commission / ESCO classification	Candidate dataset published by C. K. Shetty. LinkedIn dataset published by Arsh Koneru and collaborators
Privacy	Email addresses and phone numbers are excluded from the final parsing schema. Public examples use placeholder or redacted information. Retained fields such as company,	No personal data in ESCO	Selected candidate fields contain no direct personal identifiers. Emails, URLs, phone numbers, and contact instructions found inside job

	institution, location, and professional links are stored using field-level encryption		descriptions are removed during text cleaning
Data Quality	All 8,905 PDFs were validated. Category labels were standardised, exact duplicates were checked, perceptual near-duplicates were grouped, and gold annotations were schema-validated	Missing values, duplicate records, unused skills, and invalid relations checked	Title fragmentation, inconsistent experience labels, optional-field coverage, salary outliers, exact duplicates, description quality, and class imbalance were analysed and handled.
Ethics & Bias	Template, occupation, language, geographic, and resume-format biases were documented. Evaluation uses category-balanced sampling so that large categories do not dominate the results	ESCO is maintained for the European labour market and may contain Europe-oriented occupation titles, regulatory descriptions, and qualification terminology. However, its coverage of 3,039 occupations across all major ISCO groups provides a broad occupational reference. Recommendations are based on semantic profile similarity rather than nationality or location. Local title variants are evaluated separately, and low-confidence mappings are not	The candidate dataset is synthetic and IT-skewed. The job dataset is U.S.-dominated, LinkedIn-specific, and may reflect historical geographic, occupational, salary, and seniority biases

		automatically accepted.	
Reproducibility	Dataset slug, download date, preprocessing notebook, random seed, category alias mapping, file-hash method, perceptual-hash configuration, gold schema, and split labels are documented	ESCO v1.2.1, English CSV classification used	Dataset identifiers, retained columns, random seeds, text-cleaning functions, mapping rules, embedding model, tokenisation process, and split assignments are documented

Privacy Measures

The Resume Parsing pipeline excludes email addresses and phone numbers because they are not required for career recommendation or job matching. Public evaluation examples contain only redacted or placeholder information.

Retained structured fields such as company names, university names, locations, and professional links are stored using field-level encryption and are decrypted only for the authenticated owner of the resume.

Temporary rendered images are deleted after parsing. For selectable-text PDF and DOCX inputs, local PII detection can be applied before remote model processing. Image-only documents are processed using the approved multimodal pipeline under restricted project access.

Responsible Use

The system should recommend roles and rank jobs as decision-support only. It should not claim guaranteed hiring outcomes, fabricate skills, or use sensitive attributes for recommendation. The LinkedIn dataset is used for offline academic evaluation. The project does not use the dataset to bypass platform restrictions or automate access to protected pages. Any live job-discovery component will use only permitted public sources.

The LinkedIn dataset is used for offline academic analysis only. Live job discovery uses permitted public pages and does not bypass authentication, CAPTCHA, robots restrictions, or platform access controls. Each discovered job stores its source URL, retrieval timestamp, and structured fields for traceability.

5. Exploratory Data Analysis

5.1 Resume Parsing Dataset EDA

EDA item	Result
Total resumes	8,905

Pages per resume	1 for all files
Raw category folders	97
Canonical categories	43
Corrupt files	0
Image-only PDFs	8,905 — 100%
Files with extractable text	0
Exact duplicate files	0
Perceptual groups	6,611
Files participating in multi-file near-duplicate groups	3,291
Excess copies beyond one representative per group	2,294
Largest category	Human Resources — 358
Smallest category	BPO — 29
Category imbalance ratio	Approximately 12.3:1
Median image resolution	962 × 1,600 pixels
Median file size	231 KB
Median OCR-estimated length	Approximately 374 words

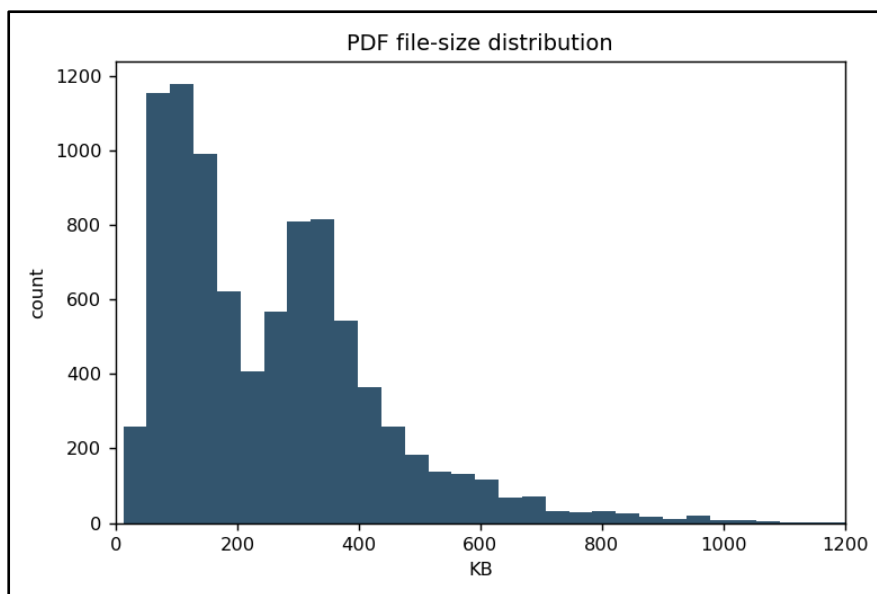


Figure 1 shows file size (in KB) distribution in resume dataset

5.1.2 Category and Label Analysis

The original archive contains 97 folder names, but many represent different spelling or formatting variants of the same occupation category. Examples include “*Data Science*”,

“DataScience”, and “data science” resumes. These labels were mapped to 43 canonical categories.

The category distribution is imbalanced. The largest category contains approximately 12.3 times as many resumes as the smallest category. To prevent majority categories from dominating evaluation, the gold set was sampled across all 43 canonical categories.

5.1.3 Duplicate Analysis

No exact byte-level duplicates were identified. However, perceptual hashing identified substantial visual duplication. The 8,905 files formed 6,611 perceptual groups. A total of 3,291 files belonged to groups containing more than one visually similar resume, representing 2,294 excess copies beyond one representative per group.

Perceptual groups were considered during gold-set selection to prevent visually similar documents from appearing in different evaluation splits.

5.1.4 Feature Relationship Analysis

Common skills such as Communication, SQL, Leadership, Project Management, Python, Microsoft Office, SQL Server, and AutoCAD occur across several occupation categories.

This shows that a candidate’s occupation cannot be identified reliably using only skill overlap. The downstream Career Recommendation module therefore uses the combined candidate profile:

Skills + Education + Experience + Projects + Job Titles

Generic skills are treated as supportive features, while specialised skills, education, experience, and job titles provide stronger occupational information.

Education and work-experience sections are not present in every resume. Therefore, the absence of a section is treated as a valid profile condition rather than automatically being treated as a parsing error.

Preliminary common skills:

- Communication — 11 resumes
- SQL — 9 resumes
- Leadership — 7 resumes
- Project Management — 7 resumes
- Python — 7 resumes
- SQL Server — 6 resumes
- Microsoft Office — 6 resumes
- AutoCAD — 6 resumes

5.1.5 Gold Annotation Field Coverage

Field group	Records containing the field
Skills	86
Education	69

Experience	80
Projects	5
Certifications	27
Job titles	85

Projects and certifications are comparatively sparse. Therefore, final evaluation should report per-field results in addition to one overall F1-score.

5.2 Career Recommendation Dataset EDA

Metric	Value
Clean occupations	3,039
Clean skills	13,939
Skills used in occupation relations	13,475
Occupation–skill relations	126,051
Essential relations	67,600
Optional relations	58,451
Unused skills	464
Occupations with zero linked skills	0
Average skills per occupation	41.48
Median skills per occupation	37
Minimum skills per occupation	7
Maximum skills per occupation	178

The dataset contains no occupation without a linked skill. Therefore, every prepared occupation has sufficient skill information for constructing an embedding context.

5.2.1 Occupational Coverage

ISCO major group	Number of occupations
Armed forces occupations	21
Managers	350
Professionals	868
Technicians and associate professionals	646
Clerical support workers	89
Services and sales workers	204
Skilled agricultural, forestry and fishery workers	44

Craft and related trades workers	394
Plant and machine operators and assemblers	347
Elementary occupations	76

Professionals and technicians form the largest groups, but the dataset also contains management, service, manufacturing, agriculture, craft, and elementary occupations. This broad coverage supports generalisation beyond software and technology roles.

5.2.3 Number of Skills per Occupation

The number of linked skills varies considerably across occupations. The average occupation contains approximately 41 skills, while the median is 37. The maximum is 178 skills.

Occupations with the highest number of linked skills include:

Occupation	Linked skills
Specialised doctor	178
Computer numerical control machine operator	158
Computer science lecturer	140
Food technologist	139
Powertrain engineer	137
Medicine lecturer	133
Computer hardware engineer	132
Biomedical scientist	127

Occupations with many skills are generally broad, multidisciplinary, technical, or professionally regulated. However, a larger skill count does not automatically mean that an occupation should receive a higher recommendation score. Otherwise, broad occupations could dominate the retrieval results.

Decision: num_skills is retained for analysis and context validation, but it is not directly used to increase the final similarity score.

5.2.4 Common Skill Analysis

Some skills are linked to a large number of occupations.

Common skill	Occupations containing the skill
Use different communication channels	340
Create solutions to problems	339
Manage staff	334

Keep records of work progress	321
Manage budgets	319
Have computer literacy	318
Quality standards	313
Troubleshoot	300

These skills are valuable across different careers but are less useful for distinguishing one occupation from another.

Decision: frequently occurring skills are retained, but they receive lower specificity weight than specialised skills. They support the overall profile match without independently controlling the recommendation.

5.2.5 Rare Skill Analysis

A total of **2,885 skills** are linked to only one occupation. Among them, **2,025 are essential skills** for their linked occupation.

Examples include specialised teaching principles, technical equipment knowledge, regulated procedures, specialised manufacturing processes, and occupation-specific tools.

Rare skills provide stronger evidence for a particular occupation because they occur in very few occupational profiles.

Decision: rare skills receive higher specificity weight and are positioned earlier in the occupation embedding context. Essential rare skills are prioritised before optional rare skills.

5.2.6 EDA Findings and Decisions

EDA finding	Decision
The dataset covers 3,039 occupations across all ISCO major groups	Use ESCO as a broad occupational reference taxonomy
ESCO terminology is mainly Europe-oriented	Evaluate local and non-European title variants separately
Some occupations contain considerably more skills than others	Do not use skill count as a direct ranking advantage
Generic skills occur across hundreds of occupations	Give common skills lower discriminatory importance
2,885 skills occur in only one occupation	Give rare skills higher specificity weight
Essential and optional relations have different meanings	Place essential skills before optional skills
Some local titles may not have an exact ESCO equivalent	Retain uncertain results as unmapped or send them for review

5.3 Job Discovery and Matching Dataset EDA

5.3.1 Candidate Dataset

Metric	Value
Total candidate records	1,000
Total features	5
Unique skill combinations	98
Unique qualifications	17
Most common qualification	Bachelors in Computer Science — 349 candidates
Experience categories	4
Most common experience level	Mid — 401 candidates
Unique job roles	22
Most common job role	Mobile Developer — 75 candidates
Missing values	0
Exact duplicate rows	0

The qualification distribution is concentrated around Bachelor's degrees in Computer Science, Information Technology, and Software Engineering. Mid- and Senior-level candidates are more common than Entry-level candidates. The job-role distribution is moderately balanced, although Mobile Developer is the most frequent class

5.3.2 LinkedIn Job Dataset Summary

Metric	Value
Total job postings	123,849
Original columns	31
Columns retained	16
Unique companies	24,428
Unique job titles	72,521
Unique locations	8,526
Exact duplicate rows	0

Job titles have extremely high cardinality, showing that they are free-text descriptions rather than standard occupational labels. Optional attributes have uneven coverage, so the ranking pipeline does not depend on any single optional field. The job title, available structured skill information, and full job description are combined into a unified job_text field.

5.3.3 Feature relationships

Skill frequency by role

The skill-role heatmap shows that common programming skills such as Python, Java, JavaScript, SQL, and C# appear across multiple roles. However, specialised combinations distinguish individual occupations:

- Frontend roles emphasise React, HTML, CSS, and JavaScript.
- Backend and Full Stack Java roles emphasise Java, SQL, and AWS.
- Data Analyst, Data Scientist, and AIML roles share Python, SQL, machine learning, and TensorFlow.
- DevOps and Kubernetes roles emphasise Docker, Kubernetes, and AWS.
- HR and Marketing contain more domain-specific skills such as recruitment.

Decision: job matching uses a combination of semantic similarity, title alignment, and experience fit rather than relying on individual skill keywords alone. The notebook explicitly added occupation-level skill analysis because aggregate skill counts were not sufficient for matching.

5.3.4 Education versus role

The candidate data shows a relationship between qualification and target role:

- Computer Science degrees occur across many software roles.
- Data Science, Cybersecurity, HR, AI, Design, and Game Development qualifications are concentrated in related roles.
- Specialised postgraduate qualifications appear in specialised occupations.

Decision: education is retained as a supporting feature. It does not override skills and experience because the candidate dataset may contain synthetic one-to-one qualification patterns.

5.3.5 Experience versus salary

Salary generally increases with experience, but Entry-level and Mid-Senior jobs contain wide salary variation and extreme outliers.

Decision: salary is not used as a core relevance signal. When used for optional filtering, salary values are log-transformed or robustly scaled, and Salary is retained as an optional attribute and represented using a separate availability indicator.

5.3.6 Experience level versus employment type

Full-time jobs dominate every experience category. The largest groups are Mid-Senior Full-time and Entry-level Full-time jobs.

Decision: employment type is used only when the candidate provides a preference. Otherwise, Full-time frequency is prevented from increasing relevance automatically.

5.3.7 Shared skills across occupations

Skill-overlap analysis shows that most roles have distinct skill sets, while related groups such as Data Analyst–Data Scientist–AIML, Frontend–Web Developer, and Backend–Full Stack–DevOps share more skills.

Decision: semantic retrieval is useful for finding related roles, but role-title and experience signals are retained to stop related yet unsuitable occupations from being ranked too highly.

6. Data Preprocessing

6.1 Resume Parsing Dataset Preprocessing

Step	Processing performed
Dataset indexing	Each PDF was assigned a file path, original folder label, canonical category, file name, and stable resume ID
File validation	PyMuPDF was used to open and render every PDF; no corrupt or unreadable files were found
Text-layer detection	PyMuPDF character extraction was used to determine whether a document contained selectable text
Category normalisation	Folder names were lowercased and cleaned, then mapped through a manually maintained alias dictionary
Exact duplicate detection	MD5 hashes were generated from the original file bytes; identical hashes were treated as exact duplicates
Near-duplicate detection	Pages were rendered as grayscale images and perceptual hashes were compared using Hamming distance
Near-duplicate criterion	Images with a pHash Hamming distance of six or less were treated as visually similar
Duplicate handling	Similar resumes were grouped into a shared dup_group; they were not deleted from the source dataset
Image rendering	PyMuPDF rendered resume pages at approximately 150–200 DPI
Image resizing	Very large images were downscaled while preserving aspect ratio
Image enhancement	Grayscale conversion, contrast normalisation, and light denoising were applied where required
OCR analysis	Tesseract OCR was used on a sample of 200 resumes for document-length and readability analysis
Gold selection	One representative per duplicate group was considered, followed by two-resume-per-category sampling
Structured annotation	Selected records were converted into fixed Candidate Profile JSON
Output normalisation	Whitespace was trimmed, empty scalar values were changed to null, and repeated skills were removed
Schema validation	All profile objects were validated using parsed_resume_schema.json
Split assignment	Development and test labels were generated with fixed random seed 42

The complete implementation is documented in `resume_dataset_pipeline.ipynb`. Exact duplicates were detected through byte-level MD5 hashing, while near-duplicates were detected using perceptual hashing. Similar documents were grouped rather than removed from the original dataset.

6.1.2 Effect of preprocessing

Metric	Raw dataset	Prepared dataset
Resume documents	8,905	8,905 retained
Folder labels	97	43 canonical categories
Corrupt files	Not validated	0
Exact duplicate files	Not identified	0
Perceptual groups	Not available	6,611
Files in near-duplicate groups	Not identified	3,291
Gold evaluation records	Not available	86
Categories represented in gold set	Not available	43
Documents discarded during category mapping	—	0

6.1.3 Category Mapping Trade-Off

The original dataset contained 97 folder labels. These were converted into 43 canonical categories using a manually prepared alias table.

Only spelling, formatting, abbreviation, or plurality variants representing the same occupation were merged. Distinct but related categories, such as Designer and Web Designing, were kept separate.

6.1.4 Model-specific Preparation

The Resume Parsing module uses a pretrained multimodal instruction-following model in a zero-shot, schema-constrained extraction setup. The model receives the resume image and returns structured JSON.

This module uses Gemma 3 as a module-specific multimodal parser, while the Career Recommendation and Job Matching modules may use different text or embedding models. The choice is justified because the Resume Parsing dataset is entirely image-based.

6.2 Career Recommendation Dataset Preprocessing

Step	Description
Column selection	Kept only useful columns such as occupation title, skill title, description, concept URI, relation type
Text cleaning	Removed extra spaces, line breaks, and inconsistent formatting
Missing value treatment	Filled missing alternative labels with blank strings
Duplicate removal	Removed duplicate occupations and skills using concept URI
Relation validation	Checked whether each occupation-skill relation points to a valid occupation and valid skill
Skill frequency calculation	Counted how often each skill appears across occupations
Specificity ordering	Ordered skills so more specific skills appear earlier in the embedding text

Feature engineering	Created num_skills, skills_list, and career_role_context
Final integration	Combined occupation, skill, and relation files into one final role-level dataset

6.2.1 Data Summary

Metric	Raw	Prepared	Change
Occupations	3,043	3,039	4 duplicate/invalid records removed
Skills	13,960	13,939	21 duplicate/invalid records removed
Occupation-skill relations	126,051	126,051 valid	0 removed
Final occupation records	—	3,039	One record per occupation
Unused skills	—	464	Retained in the master skill file but not used in occupation relations

6.3 Job Discovery and Matching Dataset Preprocessing

Step	Reproducible processing method
Dataset loading	Downloaded using kagglehub and loaded using pandas
Column selection	Removed URLs, platform metadata, ZIP/FIPS codes, sponsorship fields, view/application counts, duplicate payment fields, and other fields not required for matching
Candidate cleaning	Trimmed whitespace, corrected values such as Entry, to Entry, and split comma-separated skills into normalised lists
Categorical standardisation	Standardised company, experience, and work-type values and assigned Unknown where required.
Job skill-text preparation	Used available structured skill information and the full job description to create a consistent job-matching text.
Job-text creation	Combined title, available skills description, and full description into job_text
Text cleaning	Lowercased text, removed HTML, URLs, emails, repeated whitespace, and non-informative special characters
Experience alignment	Mapped candidate and job labels to the shared Entry, Junior, Mid, Senior, and Unknown scale
Title normalisation	Compared free-text job titles with the 22 candidate roles using RapidFuzz token-based similarity
ESCO crosswalk	Used exact ESCO titles, alternative titles, and token-based Jaccard similarity for ontology coverage analysis
BM25 preparation	Tokenised normalised candidate skills and job_text using rank_bm25
Embedding preparation	Generated 768-dimensional vectors using BAAI/bge-base-en-v1.5

Semantic similarity	Calculated cosine similarity between candidate and job embeddings
Salary processing	Retained salary-availability status; optional disclosed values were transformed/scaled using training data only
Freshness processing	Converted listed_time from Unix milliseconds into datetime and calculated age-based freshness
Validation	Checked identifiers, text availability, score ranges, mapping confidence, and pair counts after processing.

6.3.1 Shared Experience Scale

Raw job label	Standardized label
Internship	Entry
Entry level	Entry
Associate	Junior
Mid-Senior level	Mid
Director	Senior
Executive	Senior
Missing	Unknown

6.3.2 Role-title alignment

Candidate roles use 22 controlled labels, while job titles contain 72,521 free-text values. Direct equality matching is therefore unsuitable.

The final process is:

1. clean and normalise the title;
2. calculate token-based similarity against candidate roles;
3. store the best role and its confidence score;
4. check exact or high-confidence ESCO alternatives where available;
5. retain the original title for traceability;
6. leave uncertain mappings as Unmapped.

The notebook's preliminary Jaccard mapping used a cutoff of 0.50. Because this can produce semantically incorrect fuzzy mappings, the match rate is treated as coverage analysis rather than final mapping accuracy.

6.3.3 Near-duplicate clarification

Exact duplicate rows and repeated job IDs were checked. Repostings with small differences in title, company, location, or description may remain. During evaluation, company-title-location combinations are used to identify potentially repeated postings.

6.3.4 Preprocessing trade-offs

Decision	Benefit	Information potentially lost
Mapping Director and Executive to Senior	Creates one shared experience scale	Removes distinction between senior management levels

Mapping free-text titles to 22 roles	Enables controlled ranking and evaluation	Specialised titles may be simplified
Removing URLs and contact details	Improves privacy and text quality	Application instructions may be removed
Truncating long descriptions for embeddings	Reduces memory, latency, and context length	Requirements near the end may be omitted
Grouping qualifications	Reduces rare categories	Loses degree specialisation
Mapping unspecified experience to Unknown	Retains jobs without a standard experience label	Provides less precise experience scoring
Keeping salary optional	Reduces dependence on inconsistently available and geographically variable information	Reduces salary-based personalisation

6.3.5 Engineered Ranking Signals

Signal	Description
bm25_score	Keyword relevance between candidate skills and job text
semantic	Cosine similarity between candidate and job embeddings
exp_fit	Compatibility of candidate and job experience levels
role_match	Fuzzy similarity between candidate target role and job title
freshness_score	Preference for recently posted jobs
final_score	Preliminary weighted combination of the ranking signals

7. Dataset Integration

Module	Integration Status
Resume Parsing	Dataset preparation completed; internal category and schema integration completed
Career Recommendation	Completed for ESCO files
Job Discovery and Matching	Completed

a. Resume parsing:

No external dataset was merged with the Resume Parsing corpus. Internal integration was performed to create one consistent and reusable dataset.

The following keys and mappings were used:

Source field or key	Integrated field	Method
Raw folder name	canonical_category	Manual alias-table lookup
PDF file path	resume_id	Stable identifier generated from category and file name

MD5 hash	Exact-duplicate group	Identical hashes assigned to the same group
Perceptual hash	dup_group	Similar images assigned using pHash distance
Gold PDF path	Gold JSON record	Linked using resume_id
Parsed fields	Candidate Profile schema	Validated against the common JSON schema

Conflict resolution was handled by retaining the original folder label while using the canonical label for analysis. Ambiguous category mappings were manually reviewed. All selected gold records were checked for valid IDs, valid categories, available PDF files, schema conformance, and duplicate-group isolation.

The integrated Candidate Profile schema exposes skills, education, experience, projects, certifications, and job titles to the downstream modules. Resume dataset preparation and internal category/schema integration were completed.

b. Career Recommendation: three main ESCO files were integrated:

1. occupations_en.csv
2. skills_en.csv
3. occupationSkillRelations_en.csv

These were joined using ESCO occupation URIs and skill URIs.

Occupation records were joined with occupation-skill relations using occupationUri, while skill information was joined using skillUri. Relations referencing an invalid or unavailable occupation or skill URI were excluded during validation. Duplicate occupation-skill pairs were removed, while essential and optional relation types were retained separately. After integration, the number of unique occupations, unique skills, valid relations, occupations without skills, and unused skills was checked against the preprocessing summary.

c. Job Matching:

The candidate and job datasets do not contain a common identifier and are not merged through a direct database join. They are integrated using a query-document ranking structure:

- candidate profile = query;
- job posting = candidate document;
- candidate-job pair = ranking observation;
- relevance score = estimated suitability of the job for the candidate.

Schema alignment was performed as follows:

Candidate field	Job field	Integration method
skills	title + skills_desc + description	BM25 and embedding similarity

experience_level	formatted_experience_level	Shared experience-scale mapping
job_role	title	RapidFuzz title similarity and ESCO-assisted checking
Candidate profile text	job_text	BGE embeddings and cosine similarity
Employment preference, when available	formatted_work_type	Exact preference match
Location preference, when available	location	Normalised location filter
Current date	listed_time	Freshness score

Conflict resolution

- Unspecified experience is mapped to Unknown and receives neutral scoring.
- The full job description acts as the primary text source, while structured skill information is included when available.
- Conflicting work-type fields use formatted_work_type as the human-readable value.
- Free-text titles retain both the original title and mapped role.
- Low-confidence role mappings remain unmapped.
- Salary is treated as an optional preference and is not part of the core relevance score.

Integration validation

After integration, the following checks are performed:

- every candidate has a unique candidate_id;
- every job has a valid job_id;
- experience mappings contain only allowed categories;
- similarity scores remain within expected ranges;
- candidate-job pairs contain no missing IDs;
- original titles and mapped roles are both retained;
- candidate and job split assignments do not overlap where supervised learning is used.

8. Data Augmentation

8.1 Resume Parsing Module

No augmentation was applied to the original development or test resumes because evaluation must measure performance on unchanged documents.

A separate robustness dataset was created using controlled transformations:

- small image rotations;
- moderate Gaussian blur;
- JPEG compression;

- contrast reduction;
- resolution reduction;
- light image noise.

Each augmented document retains the same category and structured annotation as its source resume. Augmented versions remain within the same split as the original file so that transformed copies cannot create leakage between development and test sets.

8.2 Job Discovery and matching module

Positive training pairs are augmented through controlled skill-synonym replacement, such as Python/Python programming, SQL/Structured Query Language, and Power BI/business-intelligence dashboards.

Skill	Variants
Python	Python programming, Python scripting, Python development
SQL	Structured Query Language, SQL queries, database queries
Machine Learning	ML, supervised learning, predictive modelling
Power BI	PowerBI, business intelligence, BI dashboards
Tableau	Tableau Desktop, Tableau visualisation, data visualisation

Two augmented versions are generated for each positive training pair. Augmentation is applied only to the training split. Validation and test data remain unchanged.

Augmentation is applied only after splitting. Each augmented pair retains the original candidate ID, job ID, relevance label, and source-pair ID. Augmented examples remain in the training split and cannot appear in validation or test data. Only manually reviewed equivalent terms are used; no new skills, experience, qualifications, or job requirements are created.

9. Dataset Splitting

9.1 Resume Parsing

The Resume Parsing module uses a pretrained multimodal model in a zero-shot, schema-constrained setup. No model weights are trained; therefore, a conventional 80/10/10 training split is not required.

Dataset portion	Size	Purpose
Training set	0	No model weights are trained
Development set	34	Prompt refinement, schema checks, and error analysis
Test set	52	Final held-out Precision, Recall, and F1 evaluation
Unannotated source pool	8,819	EDA, future sampling, and robustness analysis
Total gold set	86	Structured records representing all 43 categories

The approximately 40:60 development-to-test ratio was selected because prompt development requires a smaller working set, while a larger held-out test set provides a more stable final evaluation.

The combined gold set was stratified across all 43 canonical categories. Near-duplicate groups were kept within a single split, and no visually similar resume was allowed to appear across development and test sets. The fixed random seed was 42.

Test resumes were not used as prompt examples. Robustness augmentations were kept in the same split as their source resume. These controls prevent template leakage, document leakage, and evaluation contamination.

9.2 Career Recommendation

The ESCO occupation dataset is not divided into training, validation, and test sets because it functions as the complete retrieval knowledge base. The recommendation model does not train an occupation classifier using these 3,039 roles.

Generalisation is evaluated using candidate profiles rather than by removing occupations from the ESCO reference dataset. Evaluation profiles will include:

- occupation titles not copied directly from ESCO;
- alternative and paraphrased job titles;
- Indian and other non-European title variations;
- profiles containing generic and specialised skill combinations;
- profiles with incomplete skills or experience;
- candidates whose suitable occupation is related but not an exact title match.

Candidate profiles used for prompt or threshold development will remain separate from the final held-out evaluation profiles.

9.3 Job Discovery and Matching

In candidate dataset, Candidate IDs are divided before candidate-job pair construction to prevent the same candidate from appearing in multiple splits.

Split	Ratio	Candidate count from development sample of 200	Purpose
Train/Development	60%	120	Ranking development and feature preparation
Validation	20%	40	Weight, threshold, and mapping selection
Test	20%	40	Final unbiased ranking evaluation

The 60/20/20 ratio was selected because the development experiment contains only 200 candidate profiles. An 80/10/10 split would leave only 20 candidates in validation and test, producing unstable ranking metrics. Forty candidates in each evaluation split provide more reliable NDCG and Precision estimates.

A stratified development sample of 200 candidates was selected from the full set of 1,000 candidates to control the computational cost of candidate-job pair generation. The sample preserves the main job-role and experience-level distributions. The exact number

of sampled jobs and the resulting candidate-job pair count are recorded in the preprocessing notebook.

Candidate allocation is controlled using job role and experience level where category size permits. Candidate IDs do not overlap across splits, and all candidate-job pairs belonging to one candidate remain in the same split. Pair construction is performed after candidate splitting. Augmentation is applied only to training pairs, while scalers, encoders, thresholds, and ranking weights are fitted or selected using training and validation data only.

10. Challenges Encountered

Challenge	Module	Handling
All source documents are image-only PDFs	Resume Parsing	Implemented multimodal page rendering and parsing
Real users may upload text PDFs or DOCX files	Resume Parsing	Added file-type detection and separate text-extraction routes
Real resumes may contain multiple pages	Resume Parsing	Designed/Planned page-wise extraction and structured-result merging
Category imbalance of approximately 12.3:1	Resume Parsing	Used category-balanced gold sampling and per-category evaluation
Large number of visual near-duplicates	Resume Parsing	Grouped documents using perceptual hashing and prevented cross-split leakage
Template-based resumes may not represent all users	Resume Parsing	Added robustness tests and retained format bias as a documented limitation
Projects and certifications are sparse	Resume Parsing	Reported field-level metrics with support counts
Resume documents may contain PII	Resume Parsing	Excluded email and phone, redacted public examples, and encrypted retained sensitive fields
Annotation consistency	Resume Parsing	Used written guidelines, JSON-schema validation, manifest checks, and three-person quality review
Category mapping may reduce label detail	Resume Parsing	Merged only equivalent label variants, retained raw labels, and discarded no resumes
ESCO is mainly Europe-oriented	Career Recommendation	Use its broad 3,039-occupation taxonomy while separately evaluating Indian, local, and non-European title variants
Broad occupations contain very large skill sets	Career Recommendation	Retain all occupations but prevent skill count from directly increasing recommendation score
Generic skills occur across many occupations	Career Recommendation	Calculate skill frequency and give common skills lower specificity weight

Rare skills may be hidden by long skill lists	Career Recommendation	Place rare essential skills earlier in the embedding context
Local titles may not have exact ESCO equivalents	Career Recommendation	Use alternative titles and semantic descriptions; retain low-confidence mappings as unmapped
Only 0.60% of jobs explicitly mention India	Job Matching	Use dataset for general algorithm evaluation; live discovery applies candidate-location filtering
Candidate dataset is small and synthetic	Job Matching	Use it as a controlled development set and replace it with parsed profiles during deployment
Candidate roles are mainly IT-oriented	Job Matching	Restrict evaluation to aligned jobs and document the industry mismatch
Job titles are highly fragmented	Job Matching	Map titles to controlled roles and retain confidence scores
Low-confidence ESCO mappings	Job Matching	Accept exact/high-confidence mappings and leave uncertain titles unmapped
Full-time jobs dominate work-type distribution	Job Matching	Use work type only when the candidate specifies a preference
Near-duplicate repostings may remain	Job Matching	Group company-title-location combinations during evaluation
Static 26-day snapshot	Job Matching	Use only for offline evaluation; live discovery supplies current postings
No native relevance labels	Job Matching	Use weak labels for development and independent manual relevance review for final evaluation

11. Deliverable Produced

11.1 Resumes Dataset – [Click Here](#)

11.2 Career Recommendation Dataset – [Click Here](#)

11.3 Job Matching Dataset – [Click Here](#)

12. Summary and Next Steps

Milestone 2 established the data foundation for the three main project modules. The Career Recommendation dataset was prepared from ESCO occupations, skills, and occupation-skill relations. The Job Discovery and Matching datasets were prepared using structured candidate profiles and LinkedIn job postings. EDA identified unstandardized job titles, inconsistent experience labels, uneven optional-field coverage, geographic imbalance, and class imbalance. These issues were addressed through text construction, normalization, fuzzy role mapping, controlled field standardisation, deduplication, feature engineering, and candidate-job pair construction.

For Resume Parsing, 8,905 image-only PDFs were validated and mapped from 97 raw folder labels to 43 canonical categories. Exact and perceptual duplicates were analysed, and a balanced gold set of 86 resumes was prepared with structured annotations, schema validation, quality checks, and duplicate-safe development

and test splits. The prepared data is ready for field-level Precision, Recall, and F1 evaluation during Milestone 3.

For Career Recommendation, the ESCO dataset was prepared as a reference taxonomy containing 3,039 occupations, 13,939 cleaned skills, and 126,051 occupation–skill relations. Extended EDA confirmed that the dataset covers all major ISCO occupational groups and contains an average of 41.48 skills per occupation.

The analysis identified two important skill patterns. Common skills occur across hundreds of occupations and have limited discriminatory value, while 2,885 skills are linked to only one occupation and provide stronger occupational evidence. The preprocessing pipeline therefore places essential skills before optional skills and gives rare skills a higher specificity position in the occupation context.

Although ESCO is Europe-oriented, its broad occupational coverage supports general semantic recommendation. Generalisation will be evaluated using alternative titles, Indian and other non-European title variants, paraphrased profiles, and unseen skill combinations. The original ESCO dataset and prepared career dataset remain unchanged; the additional notebook cells produce EDA tables and visualisations only.

The Job Matching data is ready for preliminary BM25, embedding, and hybrid-ranking experiments. Proxy relevance labels will be used only for initial development, while a manually labelled evaluation subset will be prepared for final Milestone 3 evaluation.

13. Team Consent

Name	Roll Number	Initial
Gaurav Kumar	22f1001105	GK
Dev Gupta	22f2000888	DG
Abhinav Ohri	24f1002064	AO
Pranav N	22f2000117	PN
Praveena N	22f3001454	PN